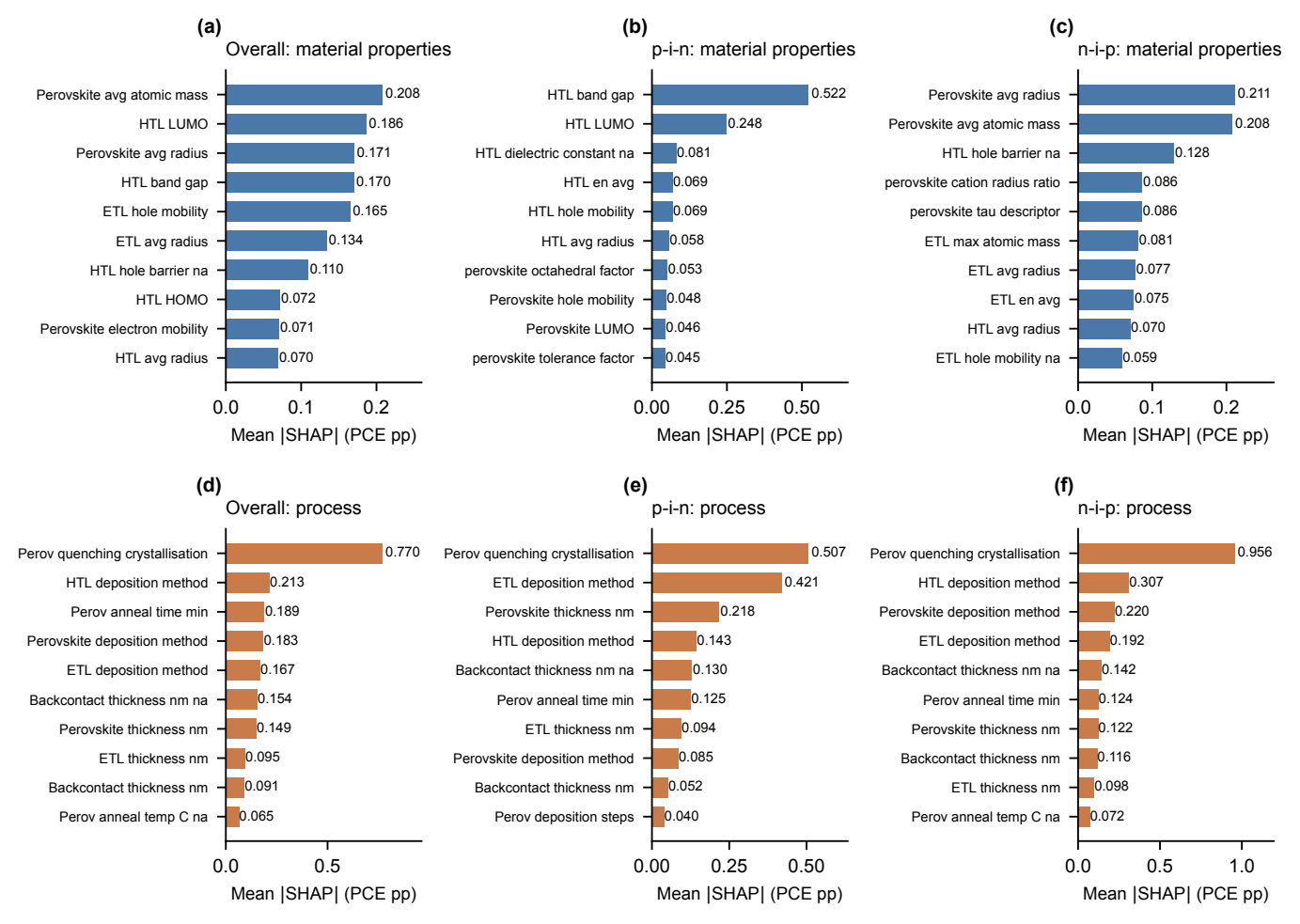
(a)
Overall: material properties
Perovskite avg atomic mass
HTL LUMO
Perovskite avg radius
HTL band gap
ETL hole mobility
ETL avg radius
HTL hole barrier na
HTL HOMO
Perovskite electron mobility
HTL avg radius
0.208
0.186
0.171
0.170
0.165
0.134
0.110
0.072
0.071
0.070
0.0
0.1
0.2
Mean |SHAP| (PCE pp)
(b)
p-i-n: material properties
HTL band gap
HTL LUMO
HTL dielectric constant na
HTL en avg
HTL hole mobility
HTL avg radius
perovskite octahedral factor
Perovskite hole mobility
Perovskite LUMO
perovskite tolerance factor
0.522
0.248
0.081
0.069
0.069
0.058
0.053
0.048
0.046
0.045
0.00
0.25
0.50
Mean |SHAP| (PCE pp)
(c)
n-i-p: material properties
Perovskite avg radius
Perovskite avg atomic mass
HTL hole barrier na
perovskite cation radius ratio
perovskite tau descriptor
ETL max atomic mass
ETL avg radius
ETL en avg
HTL avg radius
ETL hole mobility na
0.211
0.208
0.128
0.086
0.086
0.081
0.077
0.075
0.070
0.059
0.0
0.1
0.2
Mean |SHAP| (PCE pp)
(d)
Overall: process
Perov quenching crystallisation
HTL deposition method
Perov anneal time min
Perovskite deposition method
ETL deposition method
Backcontact thickness nm na
Perovskite thickness nm
ETL thickness nm
Backcontact thickness nm
Perov anneal temp C na
0.770
0.213
0.189
0.183
0.167
0.154
0.149
0.095
0.091
0.065
0.0
0.5
Mean |SHAP| (PCE pp)
(e)
p-i-n: process
Perov quenching crystallisation
ETL deposition method
Perovskite thickness nm
HTL deposition method
Backcontact thickness nm na
Perov anneal time min
ETL thickness nm
Perovskite deposition method
Backcontact thickness nm
Perov deposition steps
0.507
0.421
0.218
0.143
0.130
0.125
0.094
0.085
0.052
0.040
0.00
0.25
0.50
Mean |SHAP| (PCE pp)
(f)
n-i-p: process
Perov quenching crystallisation
HTL deposition method
Perovskite deposition method
ETL deposition method
Backcontact thickness nm na
Perov anneal time min
Perovskite thickness nm
Backcontact thickness nm
ETL thickness nm
Perov anneal temp C na
0.956
0.307
0.220
0.192
0.142
0.124
0.122
0.116
0.098
0.072
0.0
0.5
1.0
Mean |SHAP| (PCE pp)